

LAPORAN PRAKTIKUM

Aplikasi Berbasis Platform

MODUL 1 – 3 Mobile

Dart & Flutter



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PROGRAM STUDI S1 INFORMATIKA

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DASAR TEORI

GIT

Git itu semacam sistem pengontrol versi yang dipakai buat ngatur perubahan di proyek perangkat lunak. Linus Torvalds yang bikin, lho. Jadi, setiap kali ada perubahan di file, entah dikerjakan bareng-bareng atau sendirian, Git bakal nyatet semuanya. Bedanya sama sistem lain, Git ini modelnya terdistribusi. Artinya, database Git nggak cuma ada di satu tempat, tapi tersebar di banyak lokasi.

Dart

Dart adalah bahasa pemrograman yang dikembangkan oleh Google dan dirancang untuk membangun aplikasi yang cepat, modern, dan scalable. Dart bersifat object-oriented dan mendukung pemrograman berbasis class, sehingga struktur kodenya mirip dengan bahasa seperti Java atau C#. Dart banyak digunakan karena kemampuannya dalam melakukan *compile ahead-of-time (AOT)* dan *just-in-time (JIT)* yang membuat performa aplikasi menjadi lebih optimal, terutama pada pengembangan aplikasi mobile dan web. Selain itu, Dart memiliki garbage collection otomatis yang membantu mengelola memori secara efisien sehingga pengembang dapat lebih fokus pada logika aplikasi.

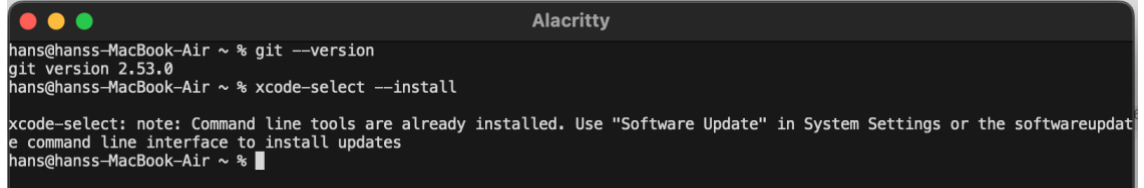
Flutter

Flutter adalah framework open-source yang dikembangkan oleh Google untuk membangun aplikasi lintas platform (cross-platform) dari satu codebase. Flutter menggunakan bahasa Dart sebagai dasar pemrogramannya dan menyediakan berbagai widget siap pakai untuk membangun antarmuka pengguna yang responsif dan menarik. Keunggulan Flutter terletak pada kemampuannya untuk menghasilkan aplikasi yang dapat berjalan di Android, iOS, web, dan desktop dengan performa mendekati native. Flutter juga menggunakan konsep *widget tree* di mana seluruh elemen UI dianggap sebagai widget, sehingga pengembangan UI menjadi lebih fleksibel, modular, dan mudah dikustomisasi.

MODUL 1 - RUNNING

1. Instalasi Flutter Environment

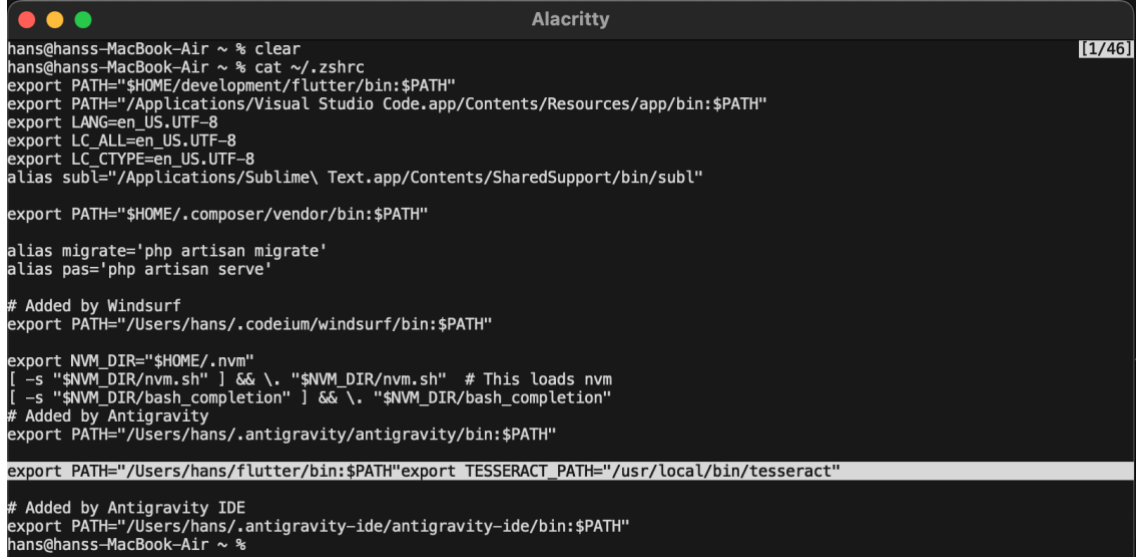
1.1. Git & Xcode Command Line (sudah terinstall)



```
hans@hanss-MacBook-Air ~ % git --version
git version 2.53.0
hans@hanss-MacBook-Air ~ % xcode-select --install

xcode-select: note: Command line tools are already installed. Use "Software Update" in System Settings or the softwareupdate
e command line interface to install updates
hans@hanss-MacBook-Air ~ %
```

1.2. Flutter SDK (sudah terinstall)



```
hans@hanss-MacBook-Air ~ % clear
hans@hanss-MacBook-Air ~ % cat ~/.zshrc [1/46]
export PATH="$HOME/development/flutter/bin:$PATH"
export PATH="/Applications/Visual Studio Code.app/Contents/Resources/app/bin:$PATH"
export LANG=en_US.UTF-8
export LC_ALL=en_US.UTF-8
export LC_CTYPE=en_US.UTF-8
alias subl="/Applications/Sublime\ Text.app/Contents/SharedSupport/bin/subl"

export PATH="$HOME/.composer/vendor/bin:$PATH"

alias migrate='php artisan migrate'
alias pas='php artisan serve'

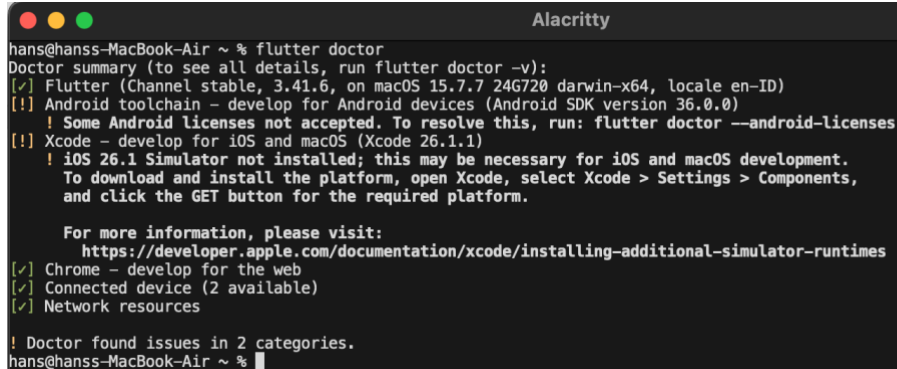
# Added by Windsurf
export PATH="/Users/hans/.codeium/windsurf/bin:$PATH"

export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh" # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion"
# Added by Antigravity
export PATH="/Users/hans/.antigravity/antigravity/bin:$PATH"

export PATH="/Users/hans/flutter/bin:$PATH"export TESSERACT_PATH="/usr/local/bin/tesseract"

# Added by Antigravity IDE
export PATH="/Users/hans/.antigravity-ide/antigravity-ide/bin:$PATH"
hans@hanss-MacBook-Air ~ %
```

1.3. Verifikasi Instalasi Flutter



```
hans@hanss-MacBook-Air ~ % flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.41.6, on macOS 15.7.7 24G720 darwin-x64, locale en-ID)
[!] Android toolchain - develop for Android devices (Android SDK version 36.0.0)
    ! Some Android licenses not accepted. To resolve this, run: flutter doctor --android-licenses
[!] Xcode - develop for iOS and macOS (Xcode 26.1.1)
    ! iOS 26.1 Simulator not installed; this may be necessary for iOS and macOS development.
      To download and install the platform, open Xcode, select Xcode > Settings > Components,
      and click the GET button for the required platform.

For more information, please visit:
  https://developer.apple.com/documentation/xcode/installing-additional-simulator-runtimes
[✓] Chrome - develop for the web
[✓] Connected device (2 available)
[✓] Network resources

! Doctor found issues in 2 categories.
hans@hanss-MacBook-Air ~ %
```

MODUL 2 - Dart

1. Dart Verification

```
hans@hanss-MacBook-Air ~ % dart --version
Dart SDK version: 3.11.4 (stable) (Tue Mar 24 01:02:20 2026 -0700) on "macos_x64"
hans@hanss-MacBook-Air ~ %
```

2. Dart Basic Syntax

```
hans@hanss-MacBook-Air /tmp % vi dart_basic.dart
hans@hanss-MacBook-Air /tmp %

void main() {
  print("===== DART BASIC =====\n");

  // 1. VARIABEL
  print("1. VARIABEL");
  String nama = 'Budi';
  int umur = 20;
  double ipk = 3.75;
  bool aktif = true;
  var jurusan = 'Informatika'; // Otomatis deteksi tipe

  print('Nama: $nama');
  print('Umur: $umur');
  print('IPK: $ipk');
  print('Aktif: $aktif');
  print('Jurusan: $jurusan\n');

  // 2. LIST (Array)
  print("2. LIST");
  List<String> buah = ['Apel', 'Jeruk', 'Mangga'];
  buah.add('Pisang'); // Tambah data
  buah.remove('Jeruk'); // Hapus data

  print('Buah: $buah');
  print('Jumlah: $buah.length\n');

  // 3. MAP (Key-Value)
  print("3. MAP");
  Map<String, dynamic> siswa = {
    'nama': 'Ani',
    'umur': 21,
    'nilai': 85,
  };

  print('Nama: $siswa['nama']');
  print('Umur: $siswa['umur']');
  print('Nilai: $siswa['nilai']\n');

  // 4. PERCABANGAN (If-else)
  print("4. PERCABANGAN");
  int nilai = 80;

  if (nilai >= 70) {
    print('Lulus');
  } else {
    print('Tidak Lulus');
  }
  print("\n");

  // 5. PERULANGAN (Loop)
  print("5. PERULANGAN");

  // For loop
  print('For loop:');
  for (int i = 1; i <= 3; i++) {
    print('Angka $i');
  }

  // For-in
  print('vFor-in:');
  for (var item in buah) {
    print('Buah: $item');
  }
  print("\n");
}
```

3. Output

```
hans@hanss-MacBook-Air /tmp % dart dart_basic.dart
===== DART BASIC =====

1. VARIABEL
Nama: Budi
Umur: 20
IPK: 3.75
Aktif: true
Jurusan: Informatika

2. LIST
Buah: [Apel, Mangga, Pisang]
Jumlah: 3

3. MAP
Nama: Ani
Umur: 21
Nilai: 85

4. PERCABANGAN
Lulus

5. PERULANGAN
For loop:
Angka 1
Angka 2
Angka 3

For-in:
Buah: Apel
Buah: Mangga
Buah: Pisang

6. FUNCTION
5 + 3 = 8
5 x 3 = 15

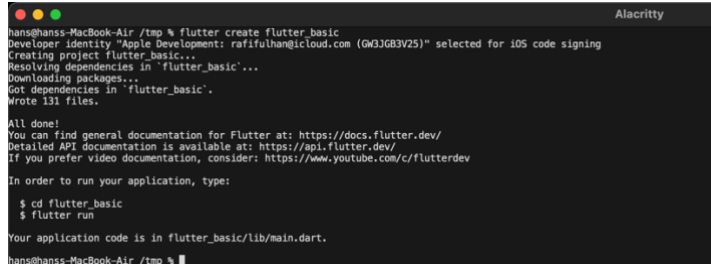
7. CLASS & OBJECT
Kucing Bersuara
Umur: 2 tahun

8. NULL SAFETY
Data: null
Data kosong
Hasil: Default

===== SELESAI =====
hans@hanss-MacBook-Air /tmp %
```

MODUL 3 – Flutter

1. Create Project Flutter



```
hans@hanss-MacBook-Air /tmp % flutter create flutter_basic
Developer identity "Apple Development: rafifulhan@icloud.com (GW3JGB3V25)" selected for iOS code signing
Creating project flutter_basic...
Resolving dependencies in 'flutter_basic'...
Downloading packages...
Got dependencies in 'flutter_basic'.
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

$ cd flutter_basic
$ flutter run

Your application code is in flutter_basic/lib/main.dart.

hans@hanss-MacBook-Air /tmp %
```

2. Program

```
3. import 'package:flutter/material.dart';
4.
5. void main() {
6.   runApp(const MyApp());
7. }
8.
9. class MyApp extends StatelessWidget {
10.   const MyApp({super.key});
11.
12.   @override
13.   Widget build(BuildContext context) {
14.     return MaterialApp(
15.       title: 'NIM & Nama',
16.       theme: ThemeData(
17.         colorScheme: ColorScheme.fromSeed(seedColor: Colors.blue),
18.         useMaterial3: true,
19.       ),
20.       home: const MyHomePage(),
21.     );
22.   }
23. }
24.
25. class MyHomePage extends StatelessWidget {
26.   const MyHomePage({super.key});
27.
28.   @override
29.   Widget build(BuildContext context) {
30.     return Scaffold(
31.       appBar: AppBar(
32.         backgroundColor:
33.           Theme.of(context).colorScheme.inversePrimary,
34.         title: const Text('Data Mahasiswa'),
35.       ),
36.       body: Center(
37.         child: Column(
38.           mainAxisAlignment: MainAxisAlignment.center,
39.           children: <Widget>[
40.             const Text(
               'NIM',
```

```

41.         style: TextStyle(fontSize: 20, fontWeight:
FontWeight.bold),
42.     ),
43.     const Text(
44.         '2311102227',
45.         style: TextStyle(fontSize: 24),
46.     ),
47.     const SizedBox(height: 30),
48.     const Text(
49.         'Nama',
50.         style: TextStyle(fontSize: 20, fontWeight:
FontWeight.bold),
51.     ),
52.     const Text(
53.         'Muhammad Rafiful Hana',
54.         style: TextStyle(fontSize: 24),
55.     ),
56. ],
57. ),
58. ),
59. );
60. }
61. }
62.

```

63. Run & Output (web device)

